Performance Improvement of ECC Scalar Multiplication

Using Windowed Non-Adjacent Form (wNAF) Method

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Problem Statement

Elliptic Curve Cryptography (ECC) is widely used in modern security systems including TLS/HTTPS, SSH, and digital signatures. The core operation in ECC is scalar multiplication (Q = k x G), and its performance directly impacts the speed of all ECC-based cryptographic operations. The standard Double-and-Add algorithm performs a point addition for roughly half of all bits in the scalar k, leading to unnecessary operations that reduce efficiency, particularly on resource-constrained devices such as IoT systems.

Objectives

- Implement the standard Double-and-Add scalar multiplication algorithm in Python.
- Propose and implement the Windowed Non-Adjacent Form (wNAF) method as an improvement.
- Benchmark both algorithms using execution time, operation count, and memory usage.
- Experimentally prove that wNAF outperforms Double-and-Add using measurable metrics.

Scope

The project will be implemented in Python 3.x using the tinyec and cryptography libraries. Benchmarks will be conducted on NIST P-256 and secp256k1 elliptic curves. Performance will be measured over 500 iterations using execution time (ms), point operation count, and memory usage (bytes).

Expected Outcome

The wNAF method is expected to reduce the number of point additions by approximately 33% compared to Double-and-Add, resulting in at least 30-40% improvement in execution time for 256-bit scalars. Security strength will remain unchanged at the 128-bit level. Results will be supported by benchmark graphs and statistical analysis.